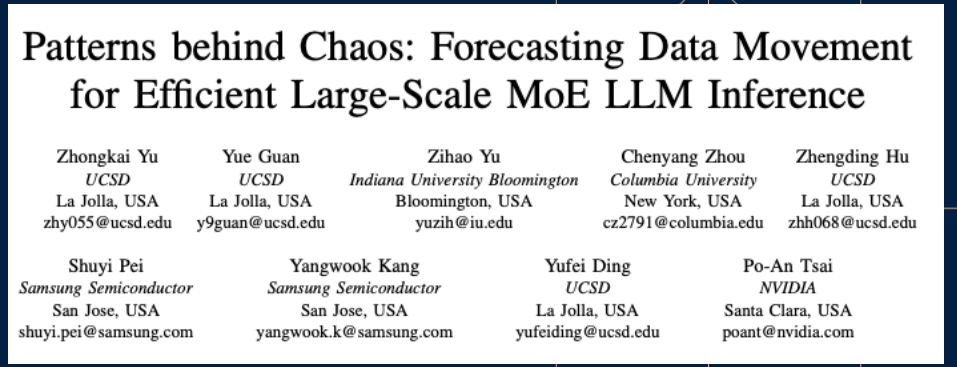
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The University of Texas at San Antonio

Patterns behind Chaos: Forecasting Data Movement for Efficient Large-Scale MoE LLM Inference

Yu, Guan, Yu, Zhou, Hu, Pei, Kang, Ding, Tsai · ISCA 2026 Best Paper

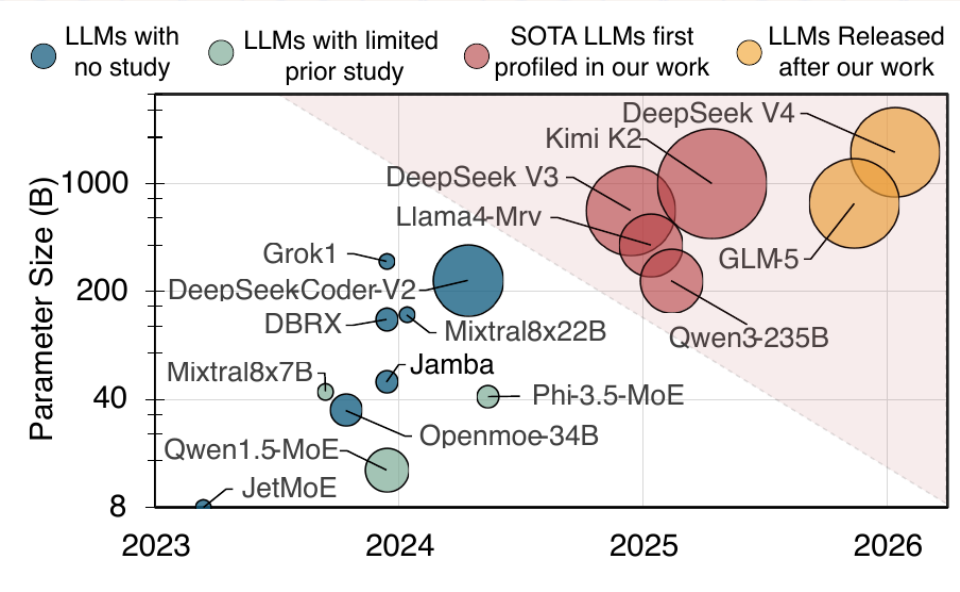
Presented by Utkarsh Kumar · 10/9/2026



Patterns behind Chaos: Forecasting Data Movement for Efficient Large-Scale MoE LLM Inference

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MoE took over the world



8→384

experts per layer

47B→1T

parameters

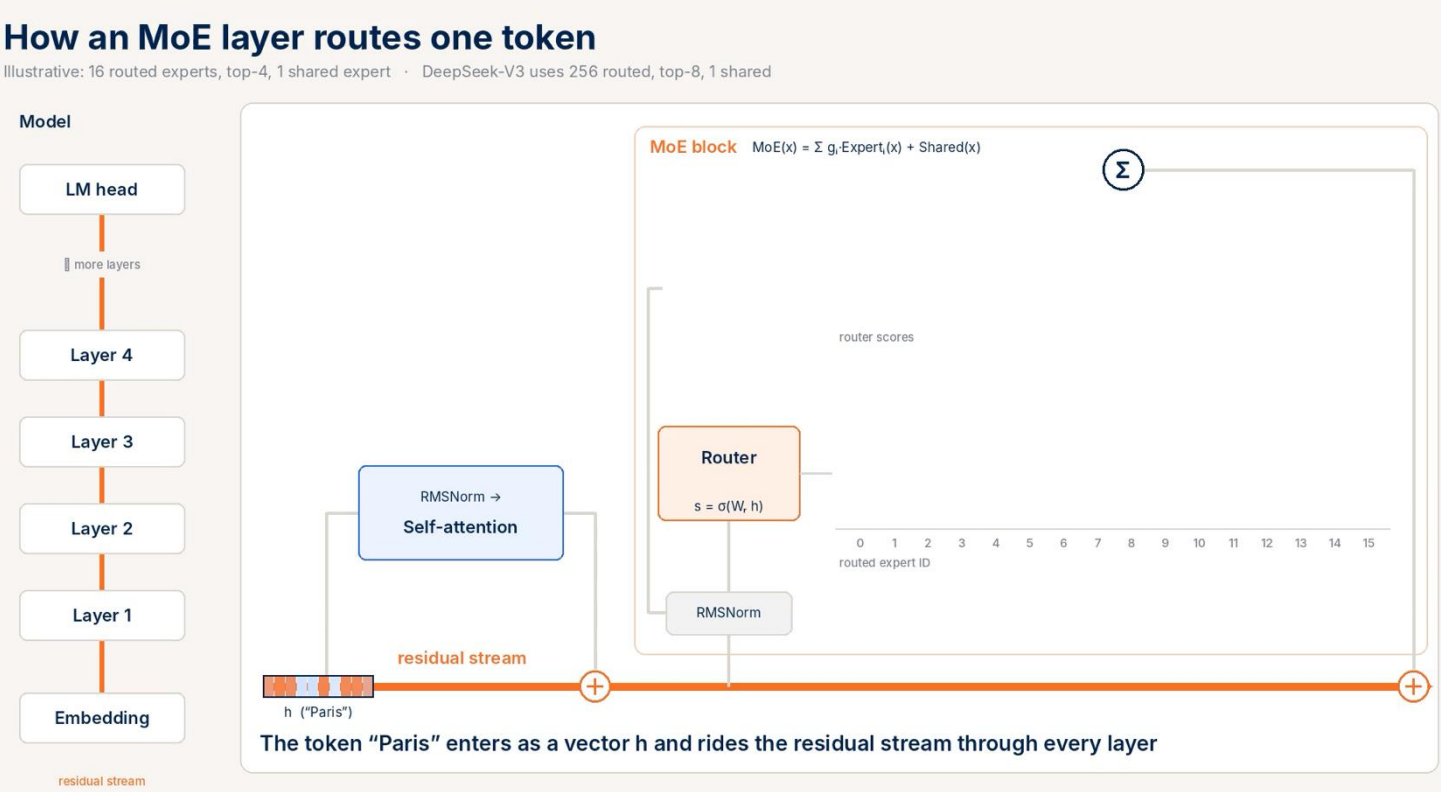
Red bubbles: DeepSeek-V3, Kimi K2, Llama 4 Maverick, Qwen3-235B, profiled here for the first time.

Prior studies looked at small models like Mixtral 8×7B.

Orange: DeepSeek-V4 and GLM-5 continue the trend.

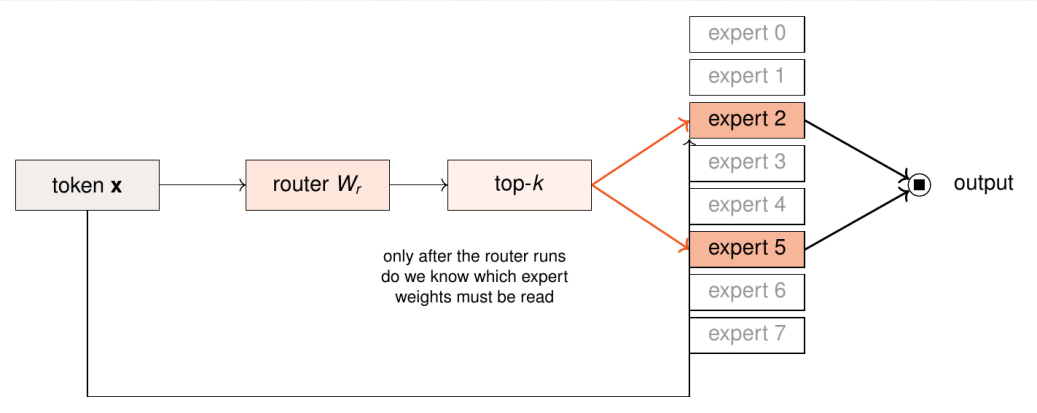
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Is Mixture-of-Experts expert selection random?



Sparse activation: big model, small cost per token

ONLY THE CHOSEN EXPERTS RUN



A router scores every expert and keeps the top k; only those experts are computed.

671B
total params
DeepSeek-V3

37B
used per token
8 of 256 experts

The catch: which 37B is unknown until the router has run, layer by layer, token by token.

If routing were really random

TWO FAILURE MODES

Temporal: nothing to prefetch

- Can't prefetch, cache or replicate experts ahead of time
- Offloading systems would migrate experts constantly

Spatial: load imbalance

- Expert load would swing with every batch
- Most GPUs idle while the busiest one finishes

$C(256, 8) \approx 4 \times 10^{14}$ possible expert sets per token per layer in DeepSeek-V3

Contributions

WHAT THE PAPER DELIVERS

4 models

24k+ requests each, the first data-movement profiling of 235B–1T MoE models

6 insights

for MoE serving-system design, from five observations

6.6×

MoE throughput on a simulated wafer-scale GPU, for 0.04% area

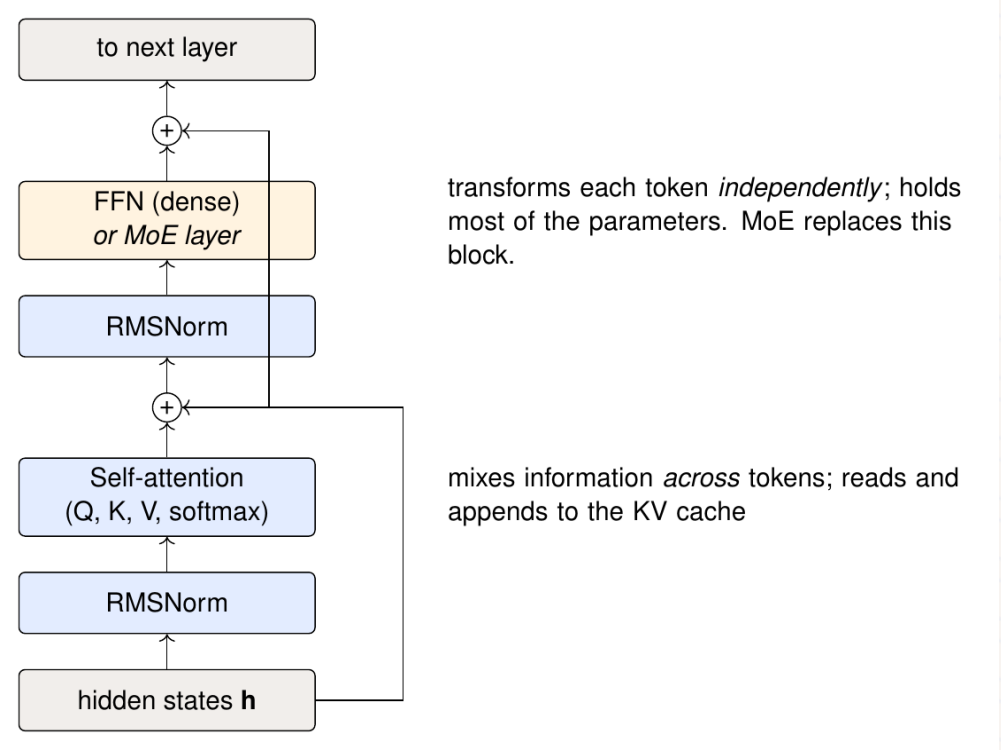
1.25×

MoE speedup on a real 8×H100 server from prefill-guided placement

Plus: 150 GB of expert-selection traces and the wafer-scale simulator, open-sourced.

The decoder-only transformer

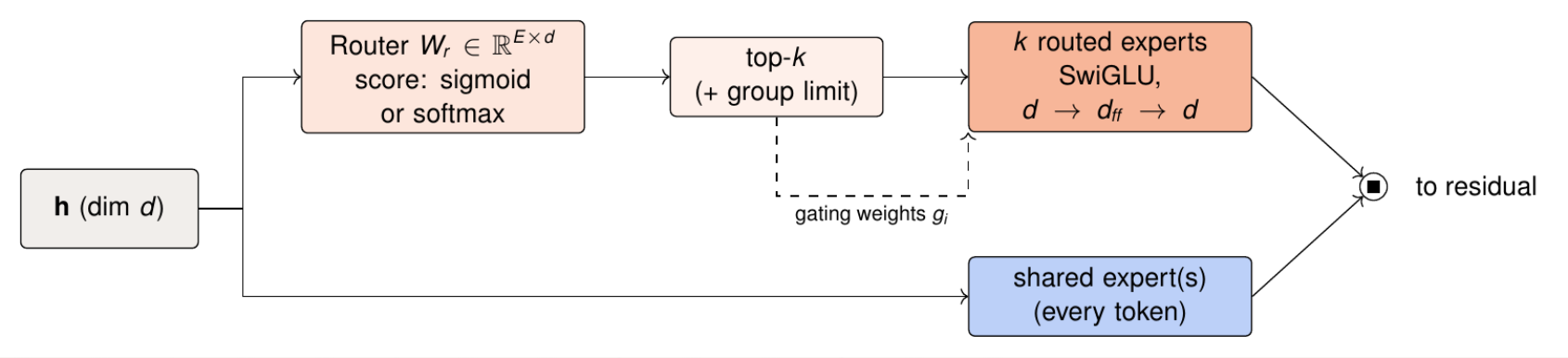
ONE LAYER, REPEATED 48–94 TIMES



- **Attention** mixes information across tokens and reads the KV cache
- **FFN** transforms each token on its own and holds most parameters
- **MoE** replaces the single FFN with many small expert FFNs

Inside an MoE layer

ROUTER, TOP-K, ROUTED EXPERTS, SHARED EXPERT



Router

One linear layer: a score per expert

Top-k

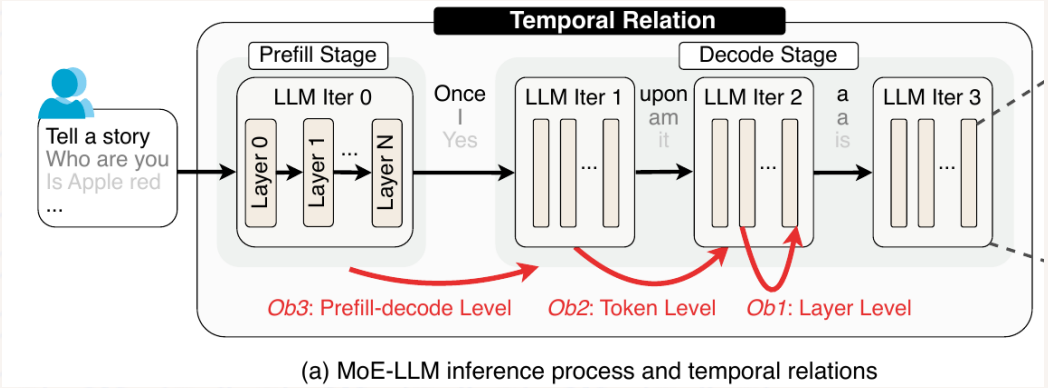
Keep the k best (k = 8; Llama 4: k = 1)

Shared expert

Every token uses it, so nothing to predict

Prefill vs. decode

TWO PHASES THAT STRESS HARDWARE IN OPPOSITE WAYS



(a) MoE-LLM inference process and temporal relations

Prefill: compute-bound

The whole prompt in one pass: one big matrix multiply per weight; sets time to first token.

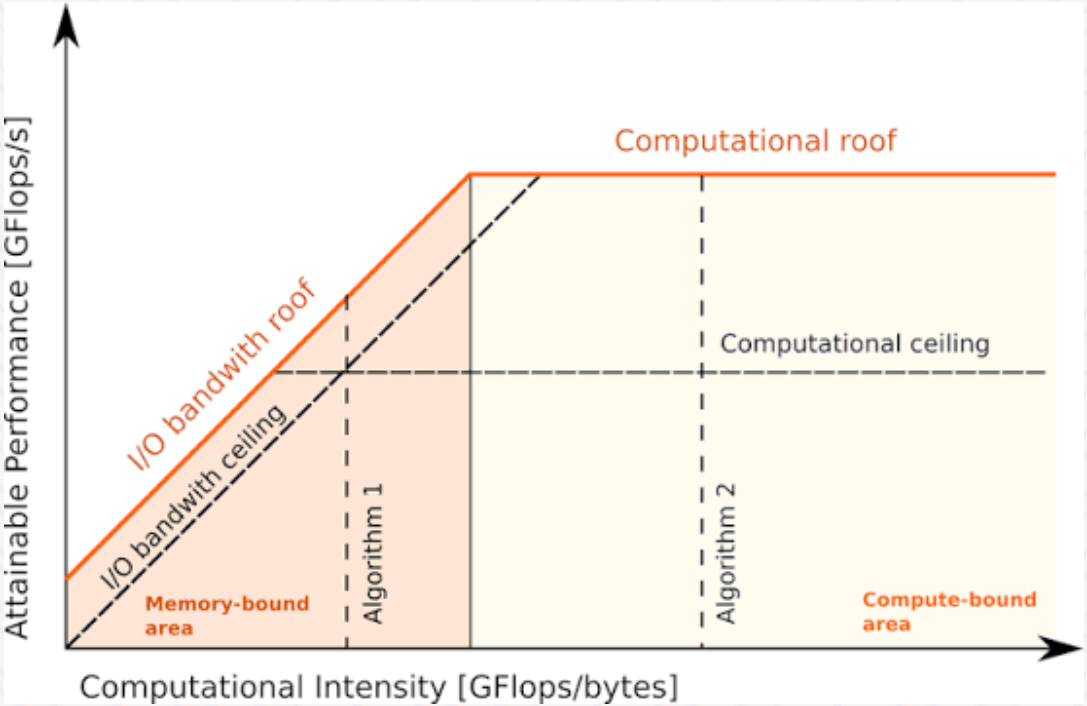
Decode: memory-bound

One token per user per step: every weight read again for only a few tokens; sets time per token.

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The roofline in one slide

IS A KERNEL LIMITED BY COMPUTE OR BY BYTES?



≈295 FLOP/B

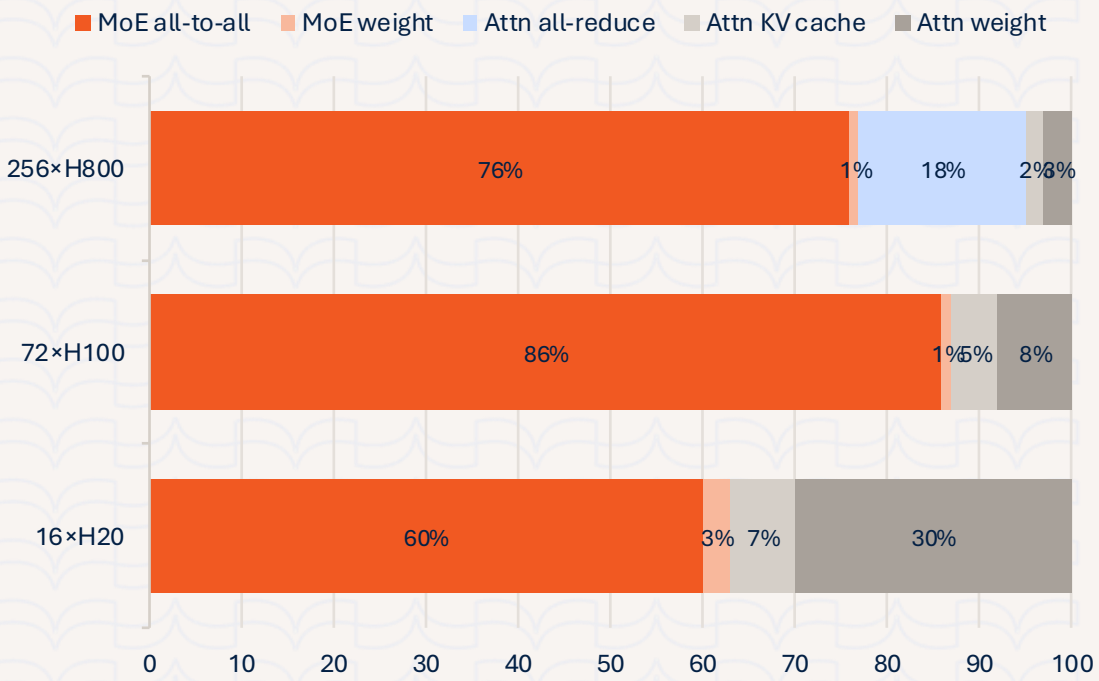
H100 ridge point

- Below the ridge: memory-bound
- Above: compute-bound
- **In decode,** intensity ≈ tokens that share each weight read
- Which of the Algo1 or Algo2 do you think is MoE models?

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Data movement dominates MoE decode

PAPER FIG. 2: MODELLED DEEPSEEK-V3 DECODE, 4K CONTEXT



Values read off the paper's figure; the breakdown is modelled, not measured.

60–90%

of decode time is MoE data movement

Almost all of it is the token all-to-all; reading expert weights is 1–3%.

Compute does not even appear in this breakdown.

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Tokens per expert decide everything

AVERAGE EXPERT LOAD = BATCH × K / EXPERTS

Model	Experts E	Top-k	B = 4096	B = 16384
DeepSeek-V3	256	8	**128**	512
Kimi K2	384	8	**85**	341
Qwen3-235B	128	8	256	1024
Llama 4 Maverick	128	1	**32**	128

Below ≈295 tokens

an expert is memory-bound: its time is set by reading its weights, not by how many tokens it serves.

Remember this for the wafer results: splitting a memory-bound expert across dies buys nothing; splitting a compute-bound one does.

Training balances experts; real traffic doesn't

WHY SKEW SURVIVES INTO INFERENCE

During training

- Auxiliary load-balancing loss (Switch)
- Bias-based balancing (DeepSeek-V3)
- Group-limited routing: 4 of 8 groups

During inference

- Balanced only on average over training batches
- Experts specialise by task and language
- A given batch loads some experts far more

>16×

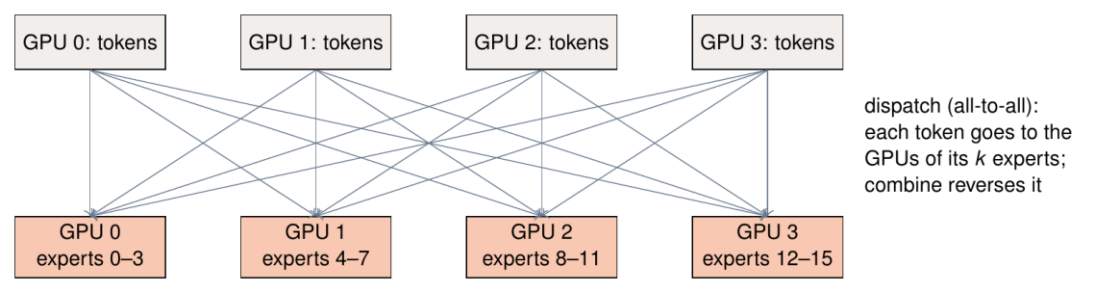
average load on the hottest experts

(Ob4)

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Expert parallelism

EXPERTS LIVE ON DIFFERENT GPUS; TOKENS TRAVEL TO THEM

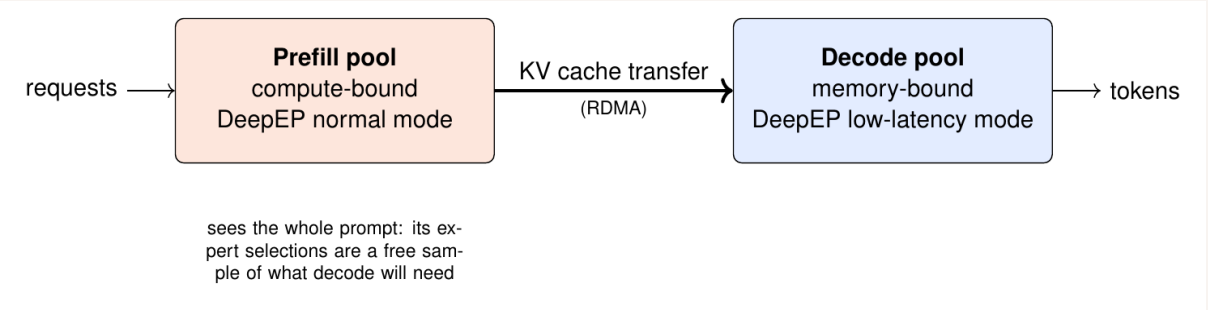


16 experts on 4 GPUs; any token may go to any GPU in every MoE layer.

- **Dispatch:** every GPU sends tokens to every other (all-to-all)
- **Compute:** each GPU runs its own experts
- **Combine:** results come back, then everyone waits for the slowest GPU

Disaggregated serving and EPLB

TWO PRODUCTION FACTS THE PAPER USES



Prefill sees the prompt first

Prefill and decode run on different machines; the KV cache is shipped across. Prefill's routing is a free sample of what decode will need.

EPLB needs history

DeepSeek's load balancer duplicates hot experts using collected decode statistics, rebalancing about every 3,000 steps.

Checkpoint

1

Decode is memory-bound

2

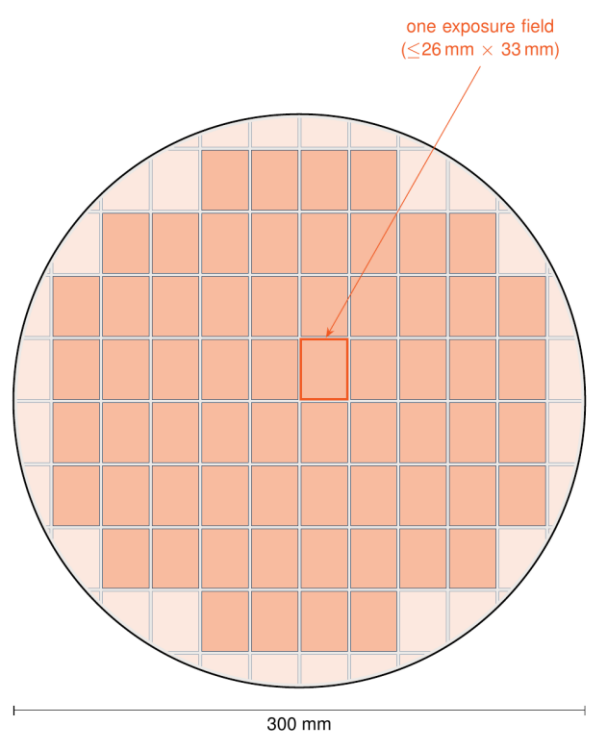
Routing is a runtime decision

3

Imbalance makes everyone wait

The reticle limit

A SCANNER PRINTS AT MOST ONE 26 × 33 MM FIELD AT A TIME



858 mm²
max single exposure

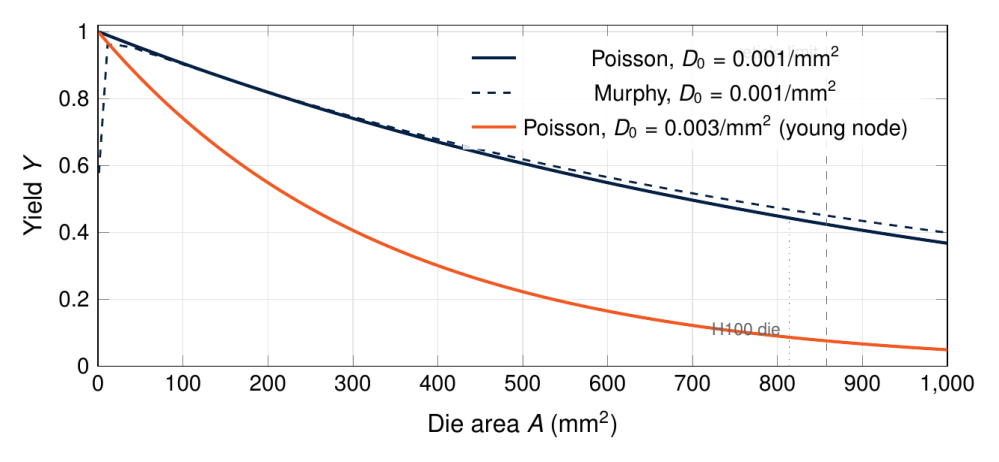
814 mm²
an H100 die

A die can't be bigger than the field the lens can image.

An H100 already fills it, so more silicon means more dies, and data crossing between them.

Yield: big dies are expensive

DEFECT-FREE PROBABILITY FALLS EXPONENTIALLY WITH AREA



≈44%

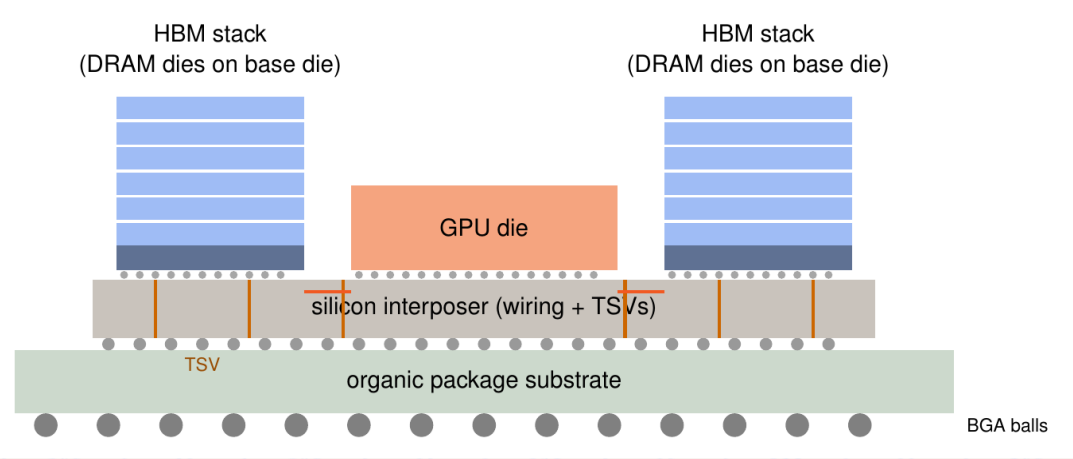
of H100-size dies are perfect

- **Harvesting:** sell with 132 of 144 SMs enabled
- **Chiplets:** test small dies, then assemble
- **Cerebras:** make each defect cost 1% of an SM

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Packaging: why HBM sits on silicon next to the die

2.5D INTEGRATION (TSMC COWOS)

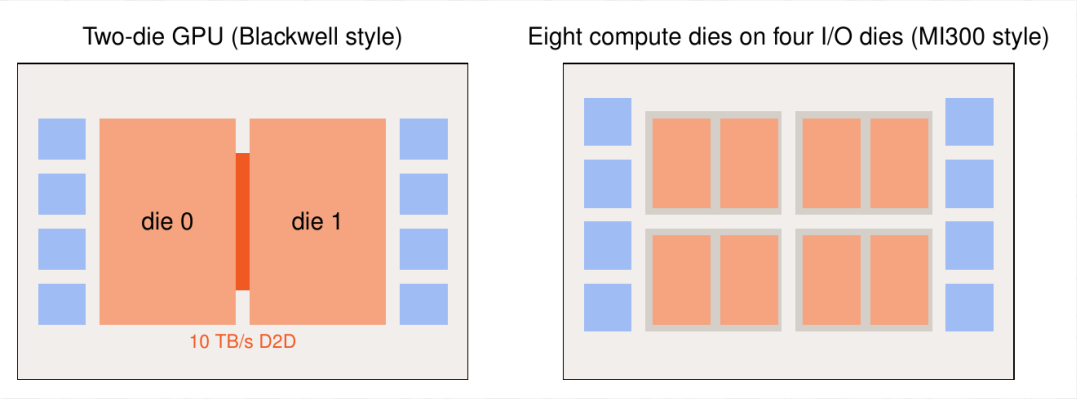


Compute and HBM dies on a silicon interposer; TSVs carry signals to the substrate.

- **HBM:** 8–16 DRAM dies stacked, a 1024-bit bus per stack
- **Only silicon** can route thousands of wires per stack
- **Dies talk** over the same bridges or short-reach SerDes

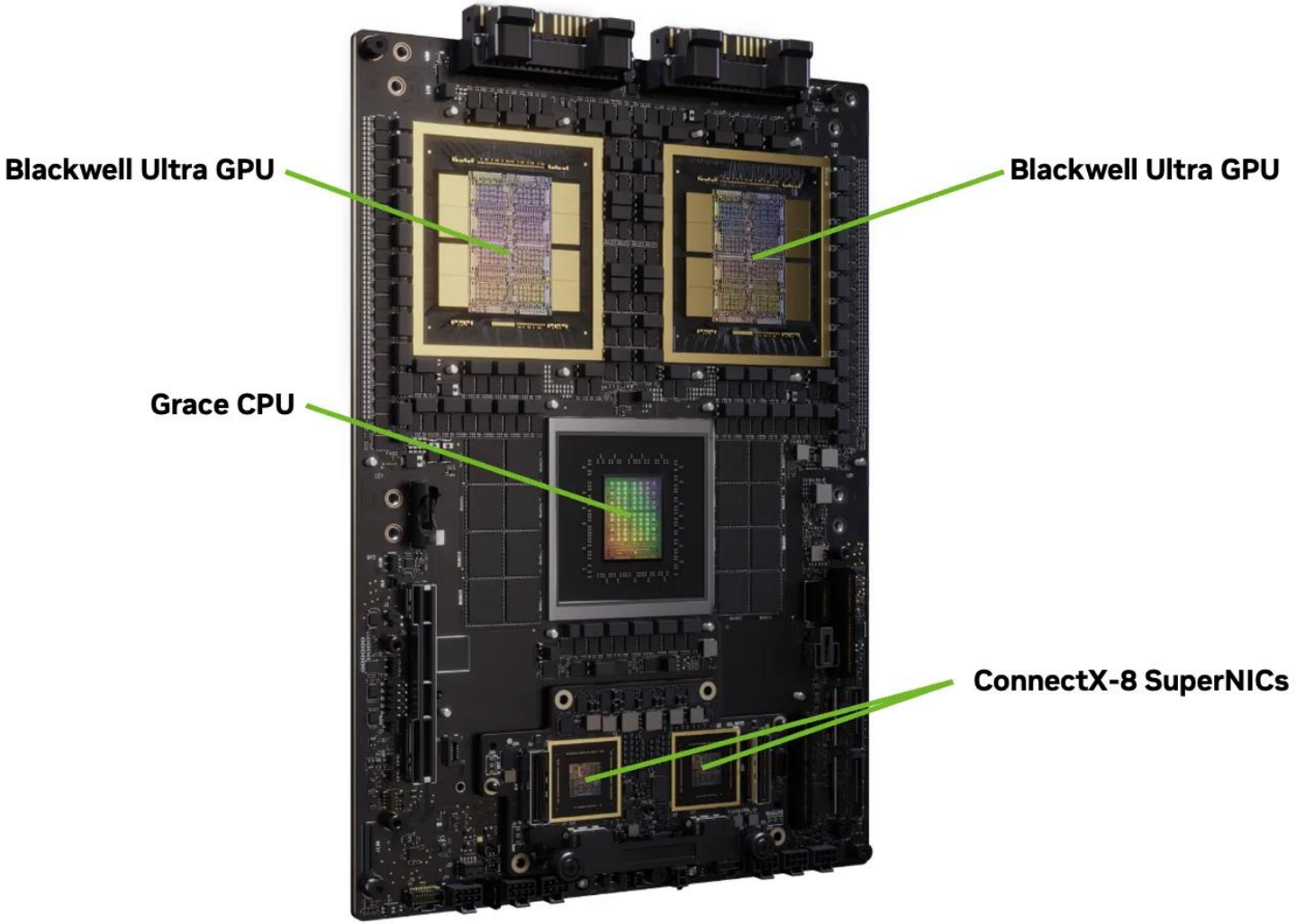
Chiplets turn a GPU into a NUMA machine

SOME MEMORY IS NEAR, SOME IS FAR



Product	Compute dies
AMD MI300X	8
NVIDIA B200	2
NVIDIA Rubin Ultra	4 (roadmap)

Hidden from software: B200 shows up as one CUDA device, so hardware must decide where data and work go.


Blackwell Ultra GPU
Blackwell Ultra GPU
Grace CPU
ConnectX-8 SuperNICs

Wafer-scale, two ways

SURVIVE DEFECTS, OR AVOID THEM

Cerebras WSE-3: one uncut wafer

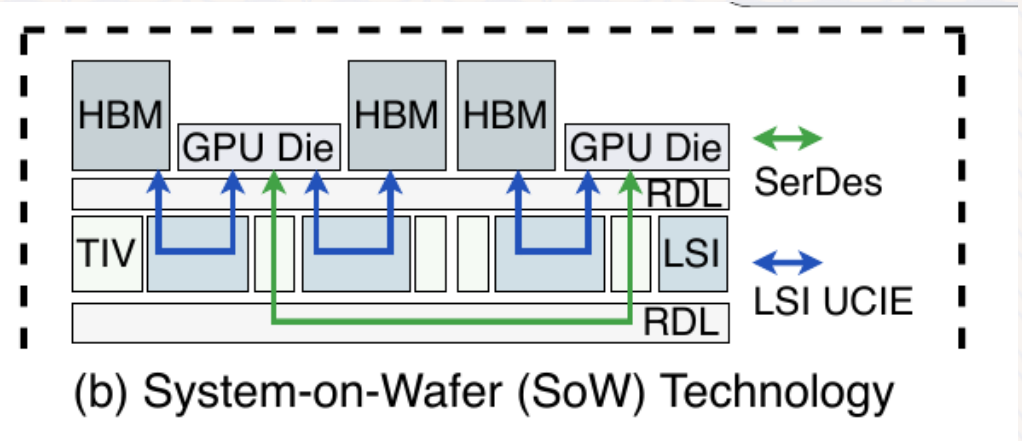
- 46,225 mm^2, 900k cores of ≈0.05 mm^2
- A defect kills ≈1% of an SM's area
- Fabric routes around dead cores
- SRAM only, no HBM

Tesla Dojo: 25 tested dies

- 25 D1 dies in a 5×5 array on InFO-SoW
- Known-good dies: bad ones never placed
- Vertical power, liquid cooled
- The paper borrows this topology

TSMC System-on-Wafer

THE MACHINE THE PAPER TARGETS

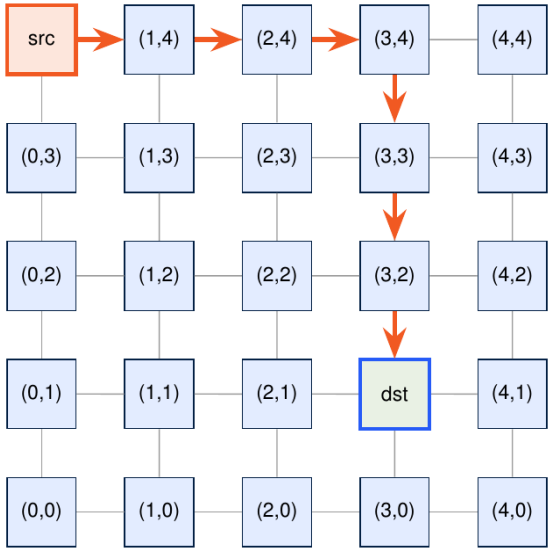


Paper Fig. 10(b)

- **24 GPU dies** + 96 HBM stacks on one wafer
- **LSI bridges** connect each die to its local HBM
- **SerDes / LSI** connect neighbouring dies
- **> 3 TB HBM,** PFLOPS of compute

On a wafer, distance is hops

A 2D MESH: DIES TALK ONLY TO THEIR NEIGHBOURS



XY route from (0,4) to (3,1):
3 hops in *x*, then 3 in *y*
Manhattan distance = 6
Latency $\approx 6 \times 200\,\text{ns}$ + HBM

Every packet from (0,4) to column 3 uses the top-row links: a hot expert causes congestion along its routes.

3.35
TB/s local HBM

1.7
TB/s per link

200 ns
per hop, shared bandwidth

XY routing: along x, then y. A hot expert's die becomes a hot spot for every route into it.

System-centric vs. model-centric

THE PAPER FLIPS THE USUAL METHODOLOGY

System-centric (prior work)

- Start from one platform: offloading, multi-GPU, PIM
- Measure MoE behaviour on that platform
- Insights may not transfer

Model-centric (this paper)

- Record the model's routing decisions themselves
- Independent of hardware
- Findings apply to any platform

Routing is a property of the model, so the GPUs used to collect traces don't change the results.

The four models

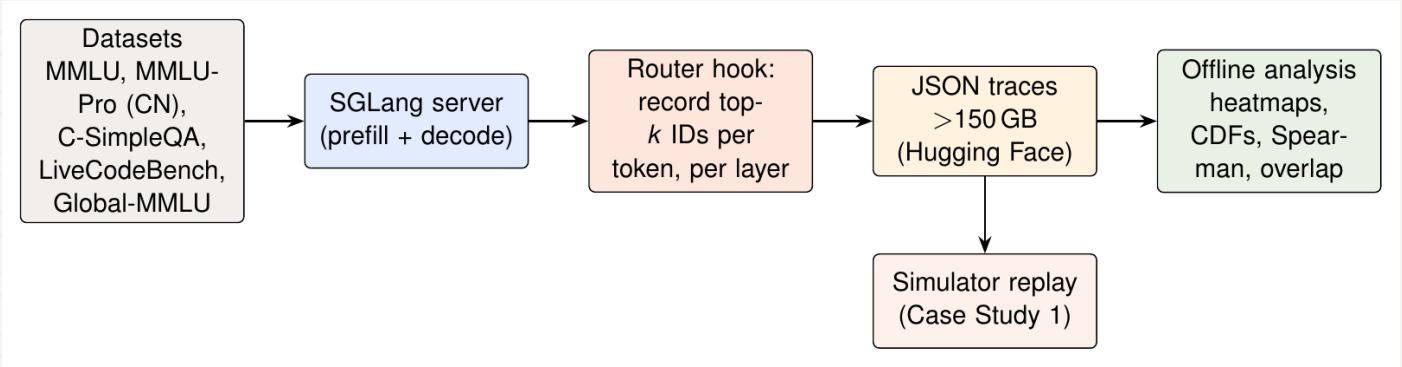
EXPERT COUNT AND TOP-K EXPLAIN MOST DIFFERENCES

	DeepSeek-V3	Kimi K2	Qwen3-235B	Llama 4 Maverick
Total / active params	671B / 37B	1T / 32B	235B / 22B	400B / 17B
MoE layers	58	60	94	24 (alternating)
Routed experts / top-k	256 / 8	384 / 8	128 / 8	128 / 1
Shared expert	1	1	0	1
Expert size (BF16)	88 MB	88 MB	38 MB	250 MB
Attention	MLA	MLA	GQA	GQA

Llama 4's top-1 routing means no expert pairs; Qwen3 has no shared expert.

The traces

EVERY ROUTING DECISION FOR EVERY TOKEN AND LAYER



24k+

requests per model

2,000+

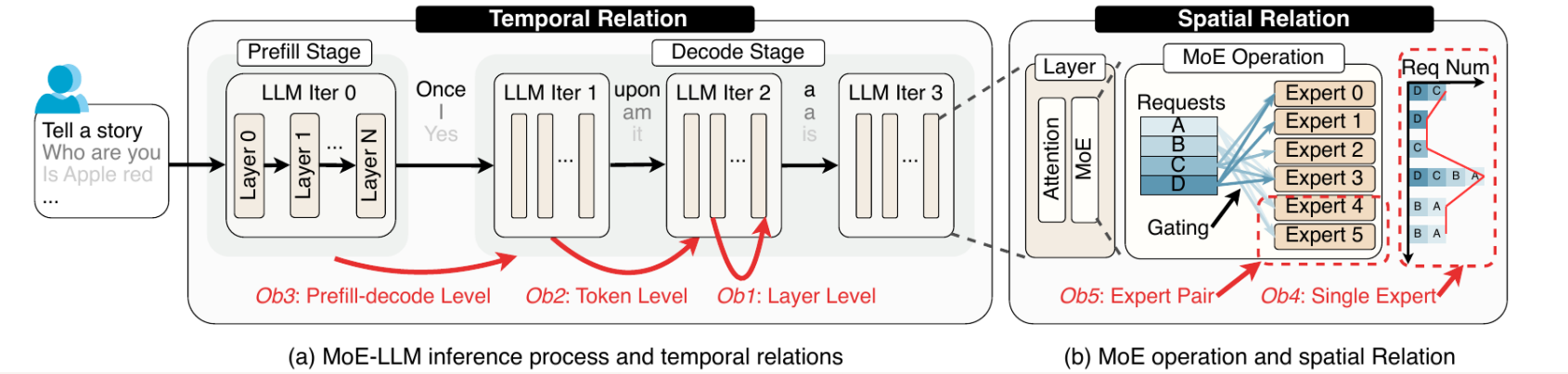
GPU-hours

150 GB

of JSON, public on Hugging Face

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The setups



(a) MoE-LLM inference process and temporal relations

(b) MoE operation and spatial Relation

Ob1 layer

next layer, same token

Ob2 token

same layer, next token

Ob3 phase

prefill → decode

Ob4 single expert

load skew

Ob5 expert pair

co-activation

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How to read the plots

CONDITIONAL HEATMAPS AND COVERAGE CURVES

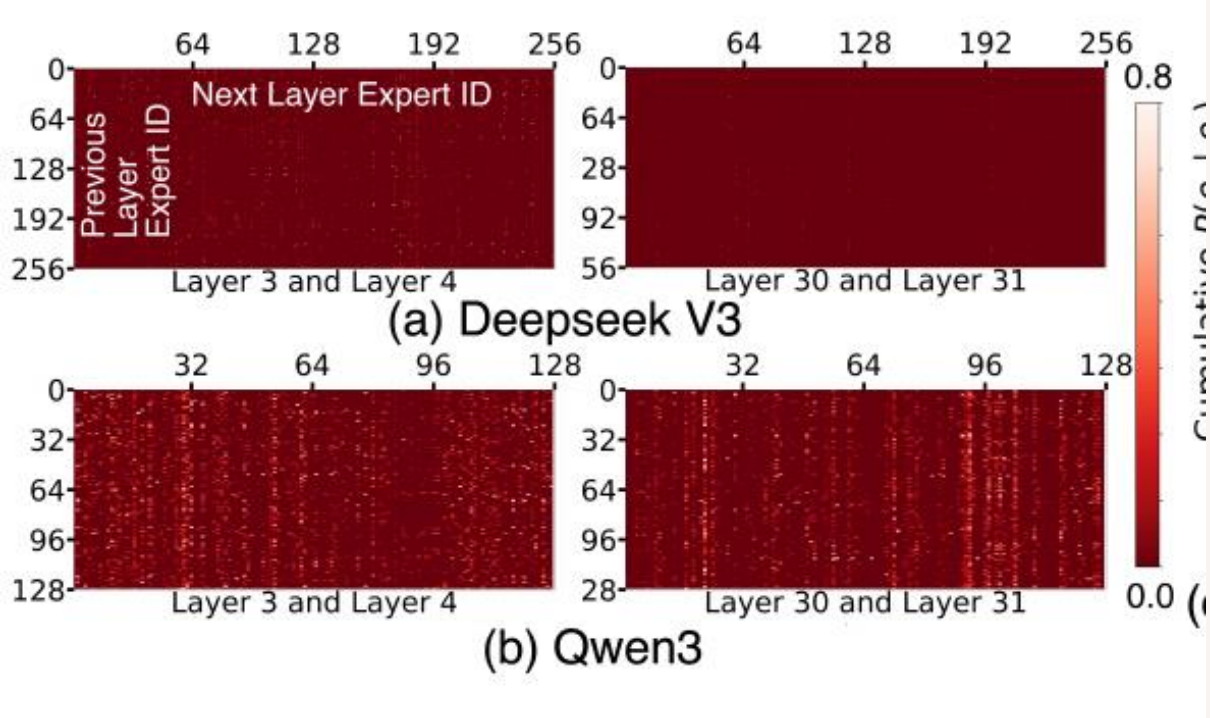


- **Each row** is P(next = j | now = i); it sums to 1
- **Sort a row and add up:** row 0 gives 0.6 from one candidate, 25% of the pool
- **Uniform routing** gives a flat row: top 25% covers 25%
- **Above the diagonal = predictable.** Bright columns = popular regardless of i

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Ob1: layer to layer

KNOWING THIS LAYER'S EXPERT NARROWS THE NEXT LAYER'S

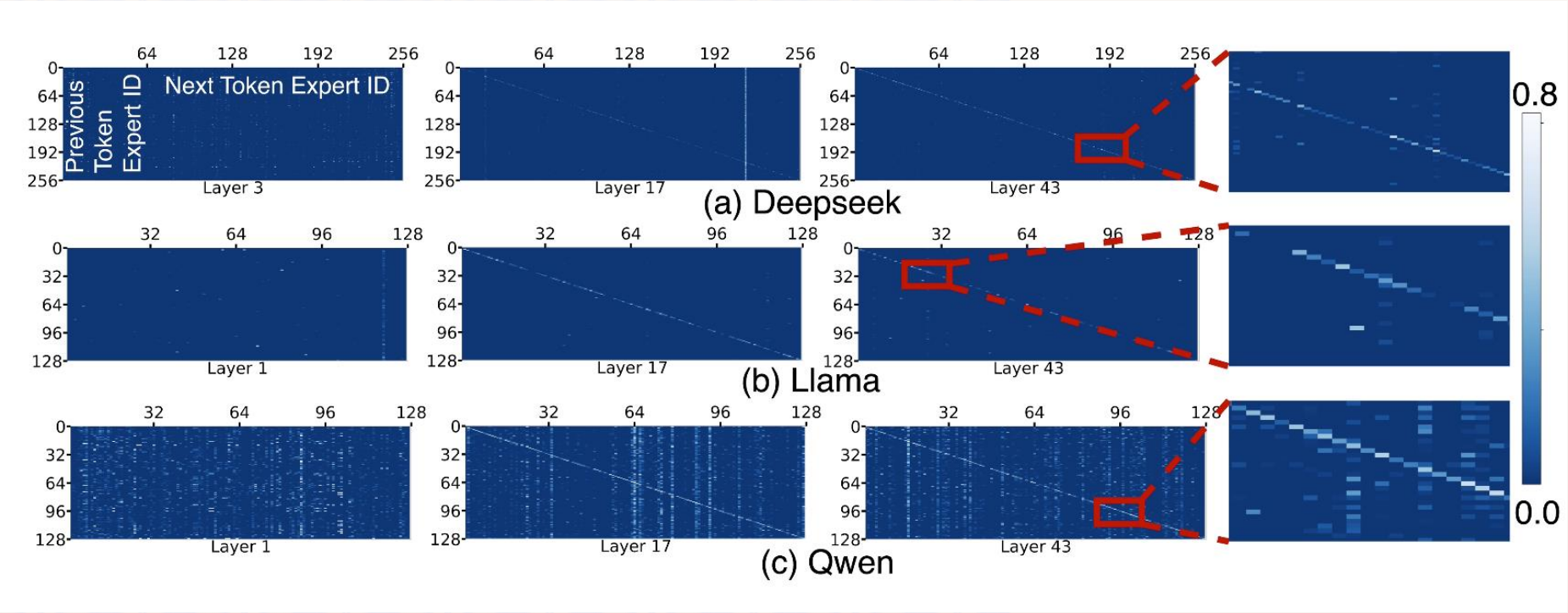


- DeepSeek-V3 :: 50%
- Qwen3 :: 65%
- Llama 4 :: 77%
- Kimi K2 :: 56%

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Ob2: token to token

DEEP LAYERS KEEP CHOOSING THE SAME EXPERT



The diagonal = same expert again for the next token.

It appears in layers 17 and 43 of every model, not in layers 1 and 3.

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Why would deep layers repeat?

OUR EXPLANATION; THE PAPER ONLY REPORTS IT

Early layers

Mostly encode the current token's identity, which changes every step, so routing changes too.

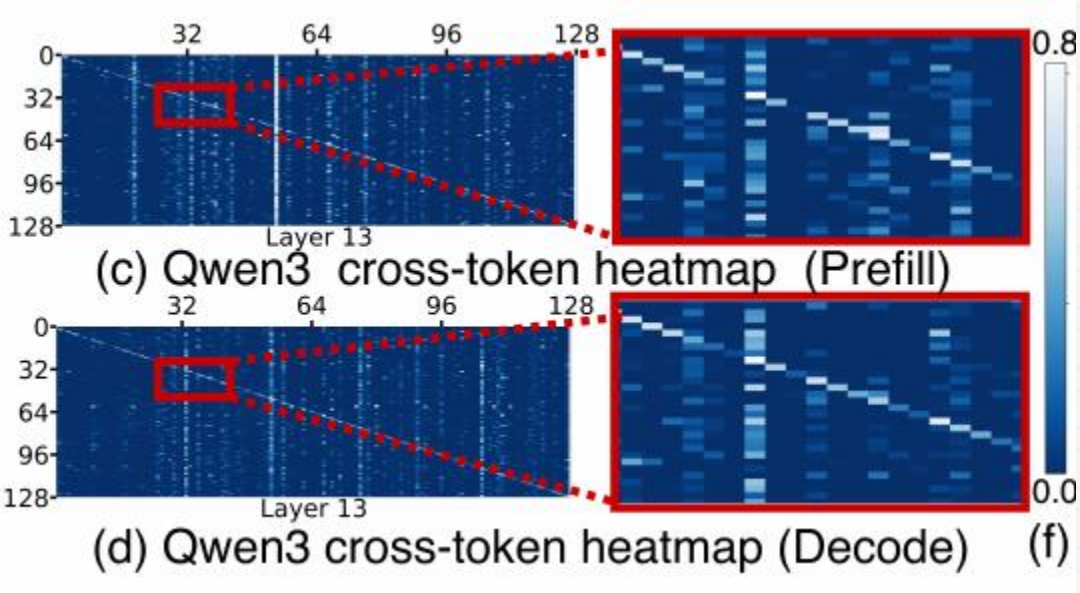
Deep layers

Encode context (topic, language, "inside a code block") that persists across many tokens, so routers repeat.

Consistent with Mixtral's report of repeated expert assignments for consecutive tokens.

Ob3: prefill looks like decode

SAME TRANSITIONS, DIFFERENT PHASES



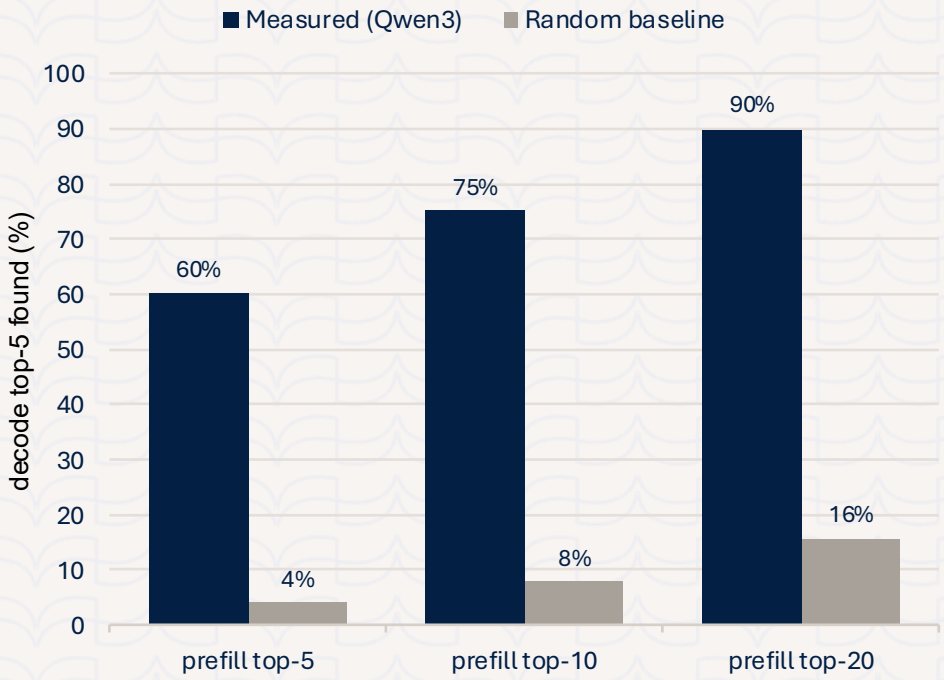
Top: Qwen3 cross-token heatmap built only from prefill tokens.

Bottom: the same, from decode tokens.

Same bright cells, same diagonal, same popular column.

Ob3 quantified

PREFILL'S HOTTEST EXPERTS ARE DECODE'S HOTTEST



Read as: of decode's 5 hottest experts, 60% are already in prefill's top 5, 90% in its top 20.

Spearman: strong for Qwen3 (0.75–0.9), moderate for DeepSeek (0.4–0.65) despite the caption's "≥ 0.7".

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Insight 1

PREFILL-DATA-DRIVEN PREDICTION

Use the expert-selection trace from prefill to predict expert selection during decode.

- Decode starts with no history; prefill offers hundreds of tokens of it for free
- In disaggregated serving, ship it with the KV cache
- Fills EPLB's cold-start gap (Case Study 2)

Insight 2: match reuse distance to memory tier

CROSS-HIERARCHY MEMORY MANAGEMENT

Layer-level (Ob1): short reuse

Needed one layer later (µs) → small, fast tier: the LLC

Token-level (Ob2): long reuse

Needed after a full forward pass (ms) → big, slower tier: local HBM

System	Fast tier	Slow tier
Multi-chiplet GPU	LLC	local HBM
CXL	local DRAM	CXL memory
SSD offload	DRAM	flash
PIM	local DRAM	remote DRAM

Case Study 1 builds only the token-level (HBM) half.

Ob4: expert load is very skewed

LLAMA 4, LAYER 7, ENGLISH MMLU



>16×

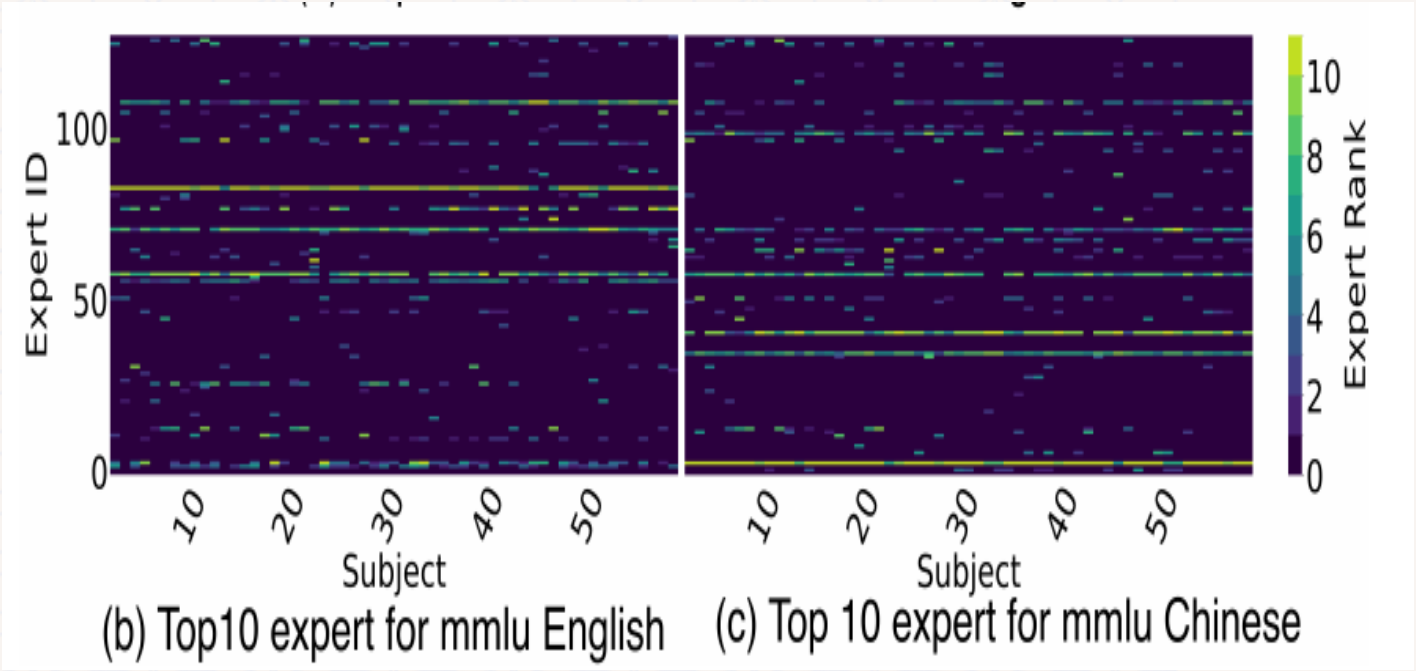
the average load

two experts in this layer

Two experts take about 16–17× their fair share; the rest form a long tail near or below average. On a GPU holding one of them, that GPU sets the layer's pace.

Ob4: task and language change the hot set

TOP-10 EXPERTS FOR EACH OF 57 MMLU SUBJECTS

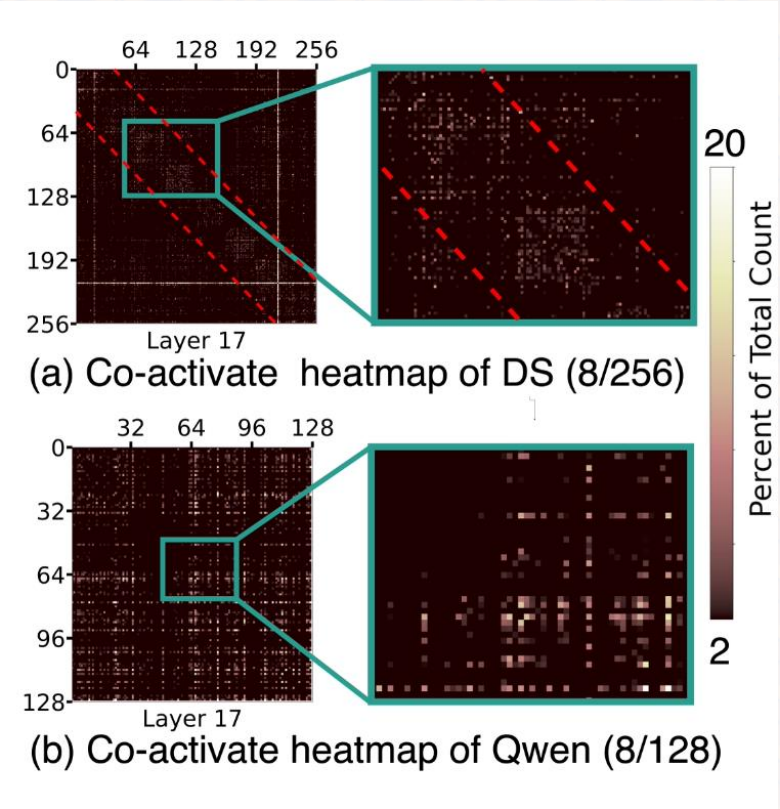


- **Horizontal lines:** experts hot for almost every subject
- **Scattered dots:** subject-specific experts
- **Same questions in Chinese:** only 2 of the 5–6 always-hot experts survive
- BTW these are **NOT** shared experts, these are despite!!!

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Ob5: experts come in pairs

CO-ACTIVATION, NORMALISED BY THE UNIFORM PAIR PROBABILITY



20–40×

chance, for bright pairs

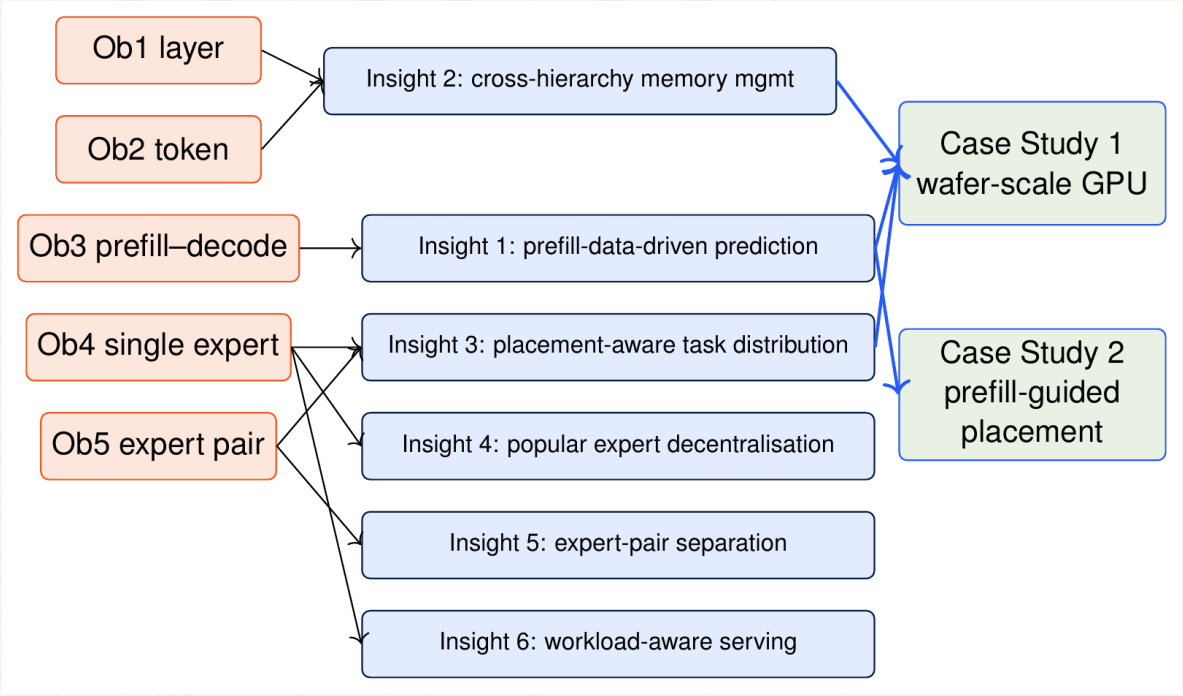
60–80%

of co-activations from top 10% of pairs

DeepSeek's diagonal band: group-limited routing makes experts with nearby IDs co-occur.

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Checkpoint: insights → case studies



Case Study 1 uses Insights 1, 2, 3. Case Study 2 uses Insight 1.

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Case Study 1

A data-movement-aware wafer-scale GPU

One GPU, or 25?

CHOOSING THE PROGRAMMING MODEL

Multi-GPU-like

- Software places data (NCCL, NVSHMEM)
- WSC-LLM, MoEntwine
- MIG disables the fast die-to-die link
- Diverges from industry practice

Single-GPU-like (this paper)

- Whole wafer is one CUDA device
- Matches Blackwell / Rubin
- Local vs. remote cost differs up to 15×
- So, hardware must handle locality

Two problems with today's GPU front end

ON A WAFER, LOCATION-BLIND SCHEDULING IS EXPENSIVE

Simplistic task allocation

The command processor hands work to any free SM, ignoring where the expert's weights are.

Flat HBM

Never keeps a local copy of a remote expert, even if it will be read again.

96%

of reads remote under random placement

24 of 25 dies hold someone else's expert

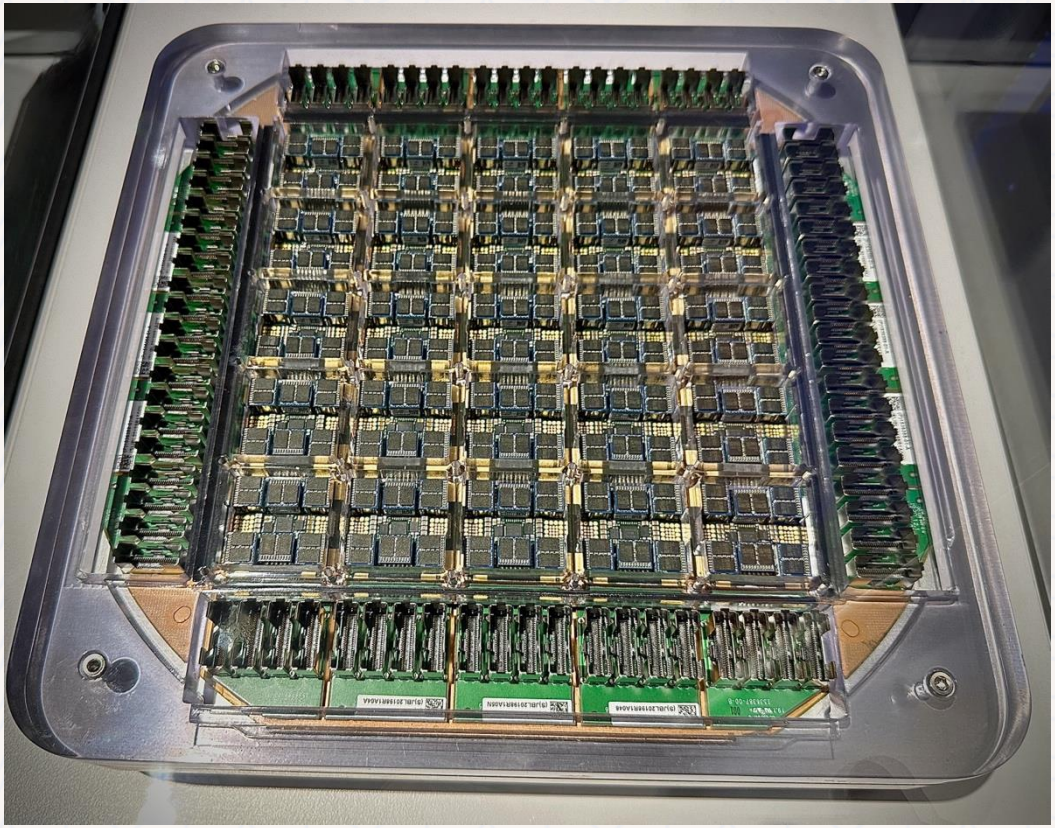
97%

of DRAM reads remote in the baseline

Qwen3, TSMC-SoW, batch 4096 (paper Fig. 14)

A toy wafer predicts the results

5 X 5 Tesla Dojo



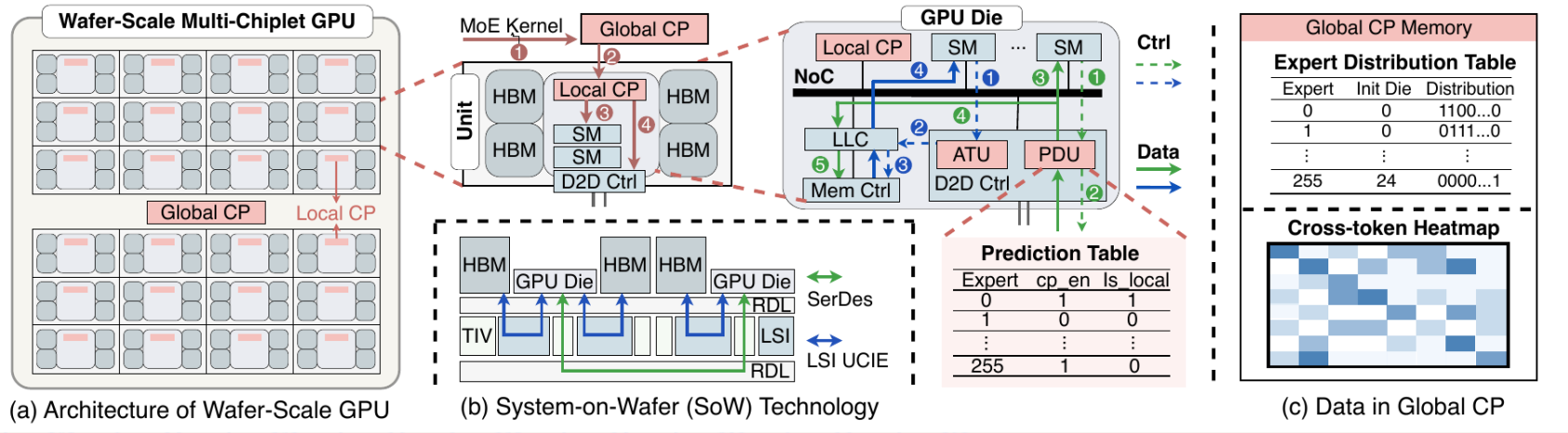
But these guys wrote their own emulators from scratch!!!!!

Also, they open-sourced it.

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Architecture

NEW UNITS IN ORANGE



Global CP
One per wafer: runs allocation and prediction, **modeled as an ARM Cortex-A76-class core**

Local CP
One per die: dispatches to its SMs
modelled as a Cortex-A72-class core

ATU
Redirects reads to local copies

PDU
Decides whether to keep remote data

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Architecture

NEW UNITS IN ORANGE

Global CP (one for the whole wafer)

- A small processor that oversees the entire wafer.
- Keeps a map of which die holds which expert.
- Keeps statistics on which experts tend to come next.
- At every MoE layer, decides which die does which work and which experts each die should keep a copy of.

Local CP (one on every die)

- A small processor that manages its own die.
- Hands that die's share of the work to its compute cores (SMs).
- Tells its die which experts to **keep copies of**.
- Reports back which copies it made.

ATU (Address Translation Unit):: Same as LUTs

- A small lookup table inside each die's link to other dies.
- When a core asks for an expert stored on another die, it checks whether a local copy already exists.
- If one does, it quietly redirects the request to local memory, so nothing has to travel across the wafer.

PDU (Prediction Unit)

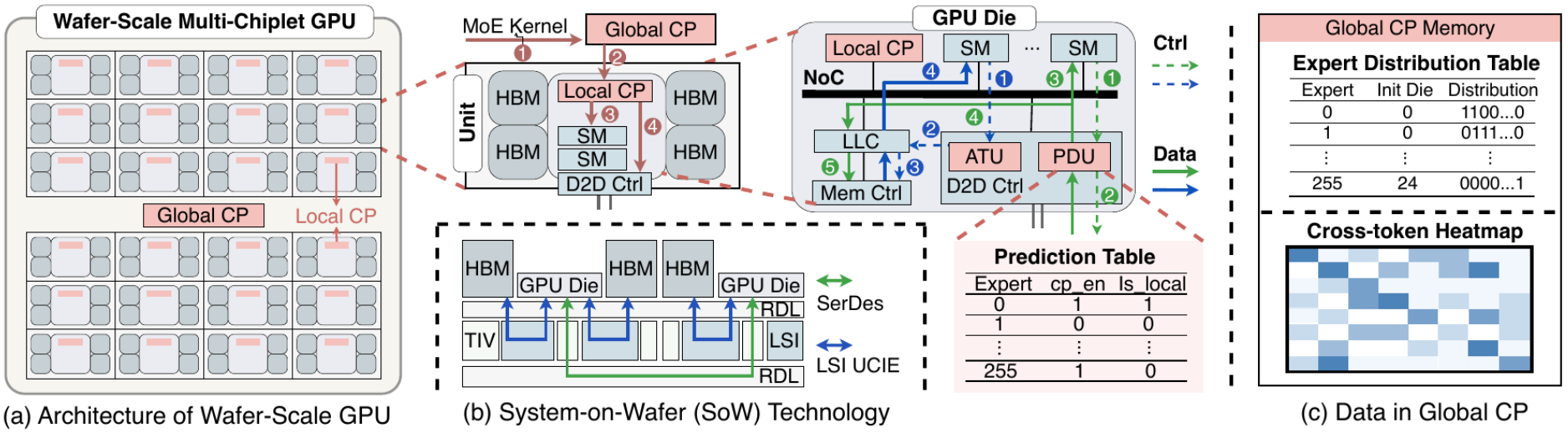
- Watches expert data as it arrives from other dies.
- Keeps two flags per expert: "keep a copy if it passes by" and "we already have a copy".
- If the first flag is on, it saves a copy into local memory for next time, without fetching anything extra.

The one real handoff between PDU and ATU: when the PDU saves a copy, it also writes the copy's local address into the ATU and sets "we have a copy". The PDU writes that entry; the ATU only reads it.

Reporting back: the Local CP collects which copies its die made and tells the Global CP, which updates its map of where every expert is. That's how the next layer's decisions know about the new copies.

Architecture

NEW UNITS IN ORANGE



Two Data Structures needed

Global CP maintains two structures. The expert distribution table stores each expert's initial die ID and distribution status as an n-bit binary code, where each bit indicates expert presence on the corresponding die. The cross-token heatmap records expert activation patterns over time, providing historical data for prediction generation.

PDU Prediction Table: Each PDU stores a prediction table with two key fields per expert: the cp en bit indicating whether the expert should be cached locally (calculated by Global CP and transferred to each die), and the is local bit tracking whether the expert is already cached in local HBM.

Task allocation (Algorithm 1)

SEND TOKENS TO THE CHEAPEST NEARBY DIE, 50 AT A TIME

Evaluation setup

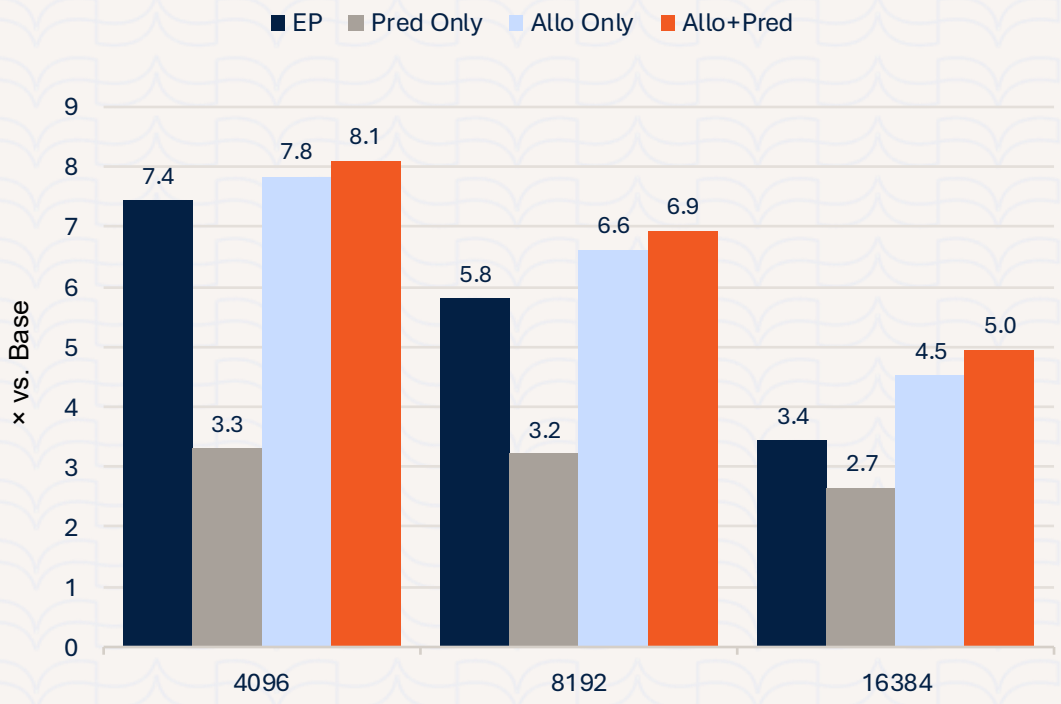
SIMULATED WAFER, REAL TRACES

Config	Mesh	HBM BW	D2D BW	FP16/die
Dojo	5×5	3.35 TB/s	1.7 TB/s	989 TF
TSMC-SoW	3×8	3.35 TB/s	1.7 TB/s	989 TF
Dojo-Enhanced	5×5	8 TB/s	2 TB/s	4,500 TF

64 MB LLC · 200 ns per hop, XY routing · local HBM 300 ns · 80 GB per die. Simulator calibrated to within 5% of a real DGX H100 (single-GPU expert GEMMs, two-GPU copies).

Headline: 6.6× over Base, 1.44× over EP at scale

MOE DECODE THROUGHPUT, MEAN OF 4 MODELS × 2 TOPOLOGIES



6.6×

Allo+Pred vs. Base

1.13×

vs. EP at 4096 tokens

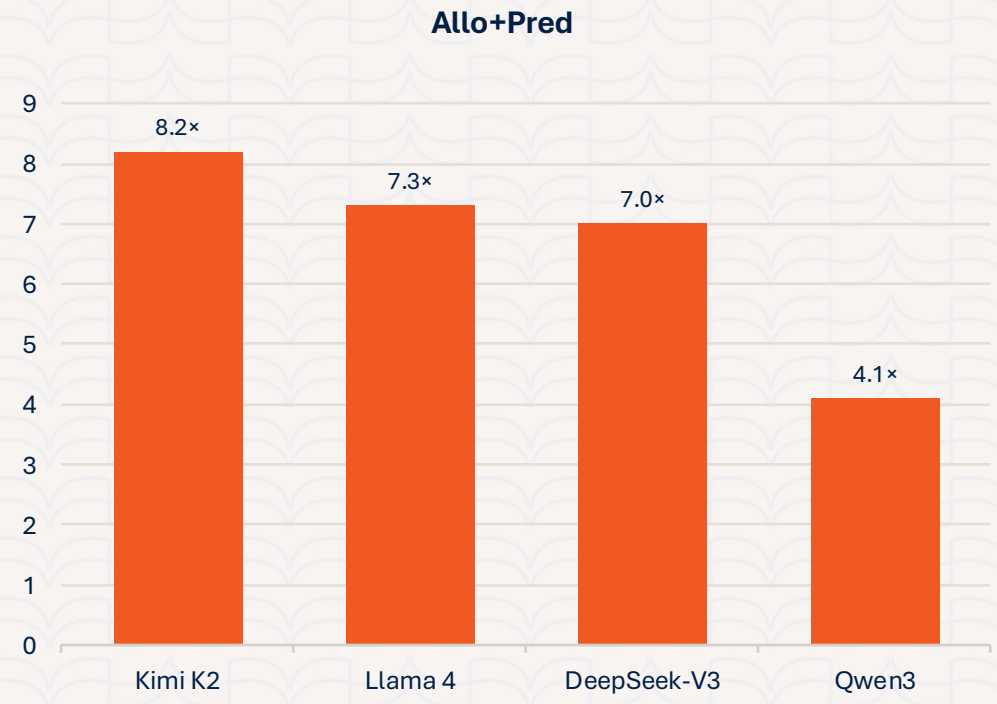
1.44×

vs. EP at 16384 tokens

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By model and by topology

ALLO+PRED VS. BASE



6.0×
5×5 Dojo

7.5×
3×8 TSMC

More experts, bigger gain: Kimi has 384.

The rectangle gains more: 3.51 vs. 3.20 mean hops, 3 vs. 5 bisection links.

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Hops and Amdahl's law

HOP REDUCTION VS. BASE, MEAN OF 8

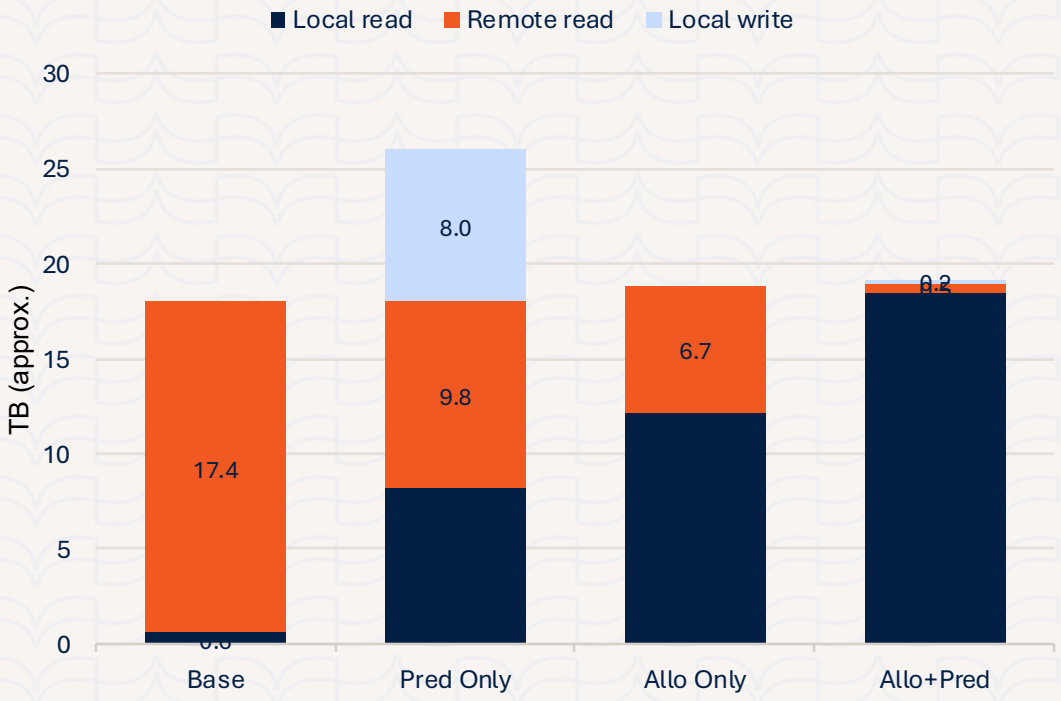
Variant	4096	8192	16384	Speedup
Pred Only	4.3×	4.6×	4.6×	3.0×
Allo Only	215×	154×	**51×**	6.3×
Allo+Pred	268×	217×	**154×**	6.6×

The two strategies are complementary: allocation spreads hot experts to neighbours, prediction lets those neighbours keep copies.

- 213× fewer hops, only 6.6× faster: compute balance is now the bottleneck
- At large batch Allo trades hops for balance (215× → 51×)
- Pred buys them back (51× → 154×)

Where the bytes come from

QWEN3, TSMC-SOW, BATCH 4096 (PAPER FIG. 14)



97% → 3%

remote share of reads

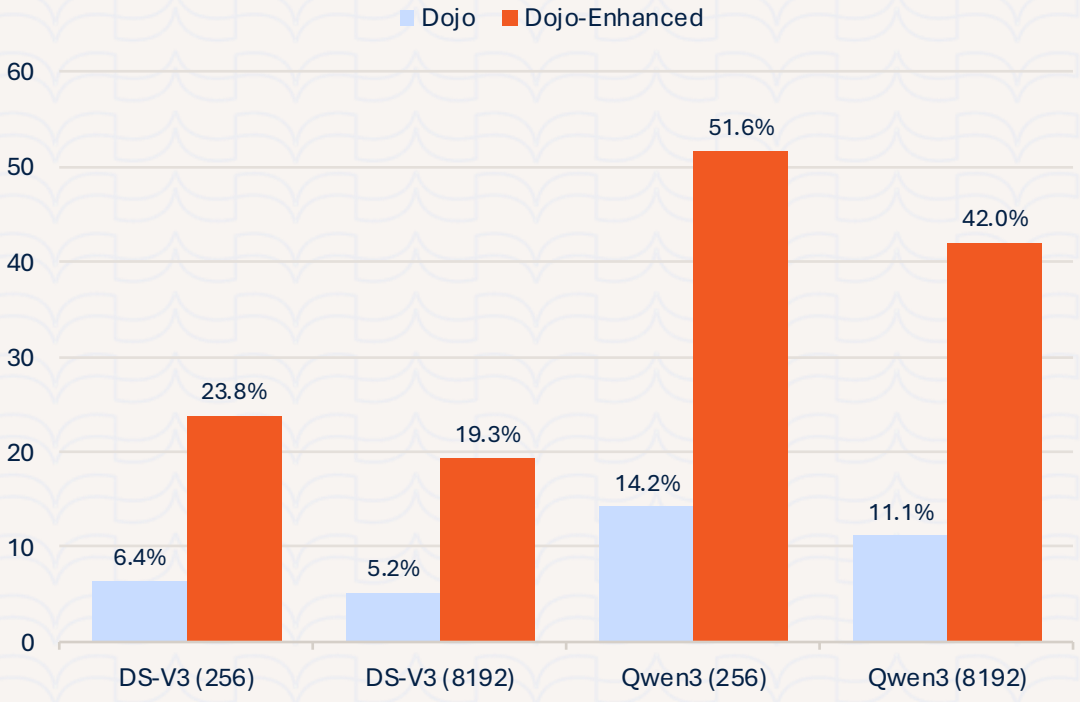
Pred Only caches a lot (≈8 TB of writes) because work lands anywhere.

Allo+Pred barely needs copies.

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Why new hardware? The host-CPU alternative

OVERHEAD OF RUNNING THE ALLOCATOR ON THE HOST (PAPER FIG. 15)



- **PCIe round trip** every MoE layer
- **Qwen worse:** 94 vs. 58 layers
- **Faster GPUs, worse:** PCIe doesn't speed up

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Cost: about 0.04% of the wafer

PAPER TABLE II

Component	Count	Area (mm^2)	Power (mW)
Prediction Table	25	0.0020	56
Address Translation Unit	25	0.0048	334
Local CP (A72-class)	25	≈7.5	≈7,000
Expert Distribution Table	1	0.0002	14
Heatmap cache (0.5 MB)	1	0.0278	185
Global CP (A76-class)	1	≈1.1	≈1,000

0.04%

area and power

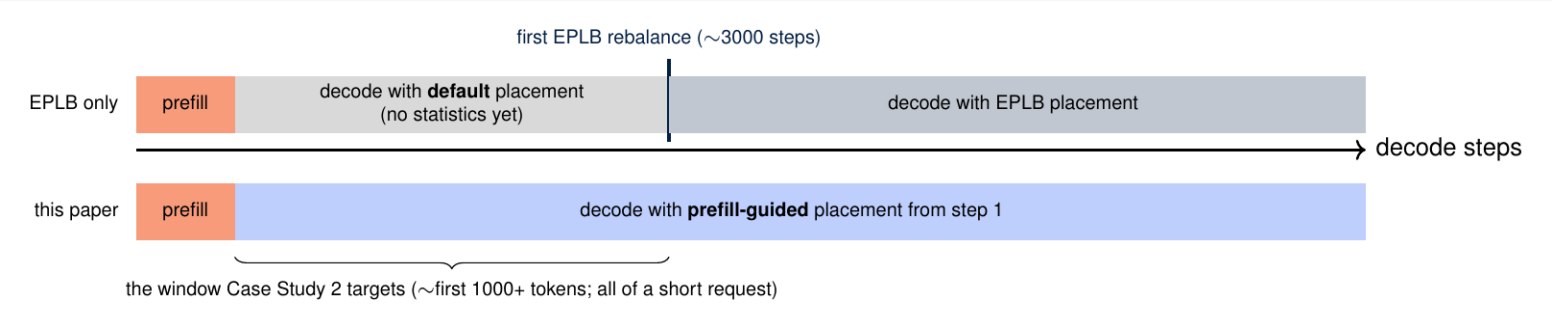
The CPU cores dominate. Area column sums to ≈8.6 mm^2, not the printed 6.13; still 0.04%.

Case Study 2

Prefill-guided expert placement on real GPUs

The cold-start problem

EPLB NEEDS DECODE HISTORY; PREFILL IS AVAILABLE IMMEDIATELY



Who runs on the default layout?

- Every new deployment's first ≈1,000+ tokens
- Every short request, entirely

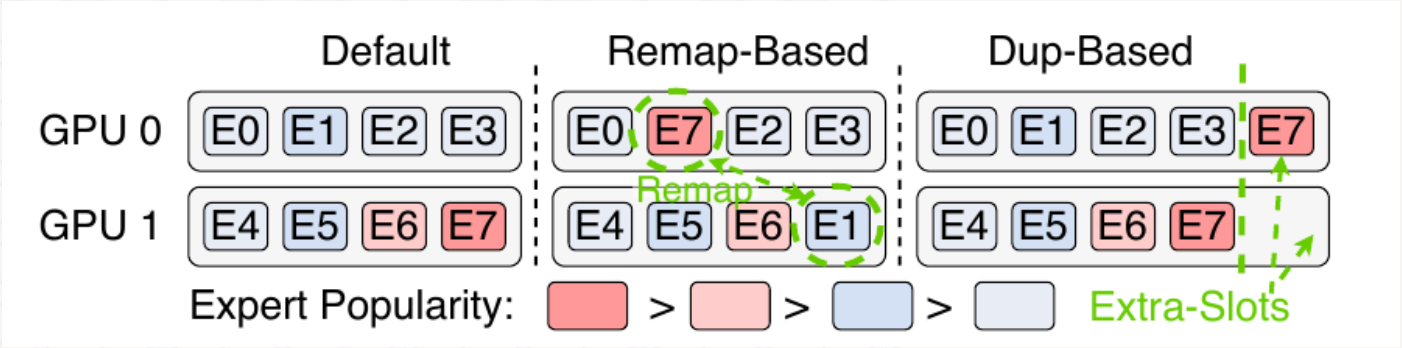
The idea (Insight 1)

Compute the decode placement from prefill's routing statistics, before decode starts.

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Two placement algorithms

PAPER FIG. 16 AND ALGORITHM 2



Remap: reshuffle

Heaviest expert first to the least-loaded GPU, 16 per GPU. No extra memory.

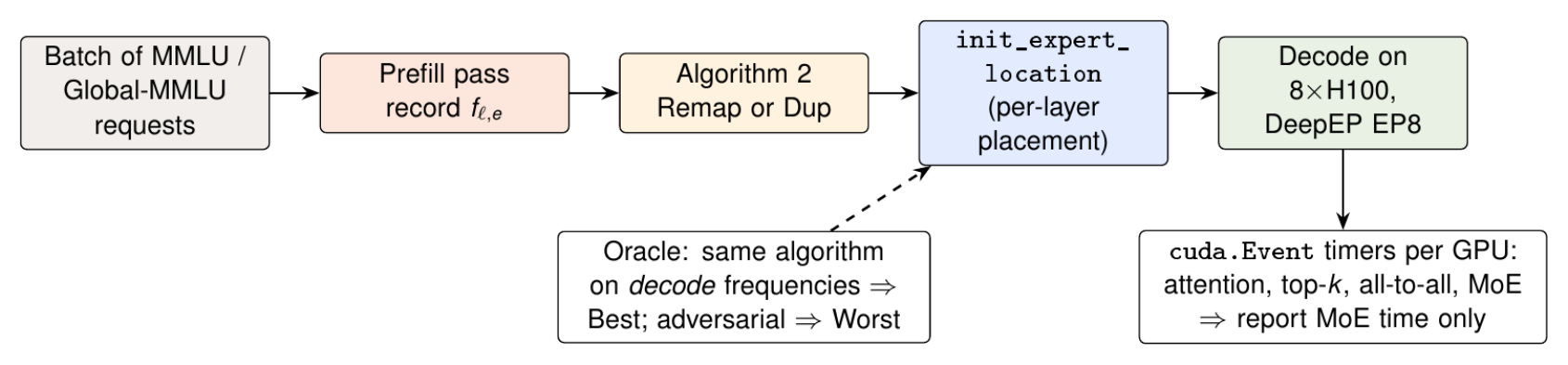
Dup: replicate

Keep the default layout; one extra slot per GPU for copies of hot experts, tokens split evenly.

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Methodology

REAL 8×H100, QWEN3-235B

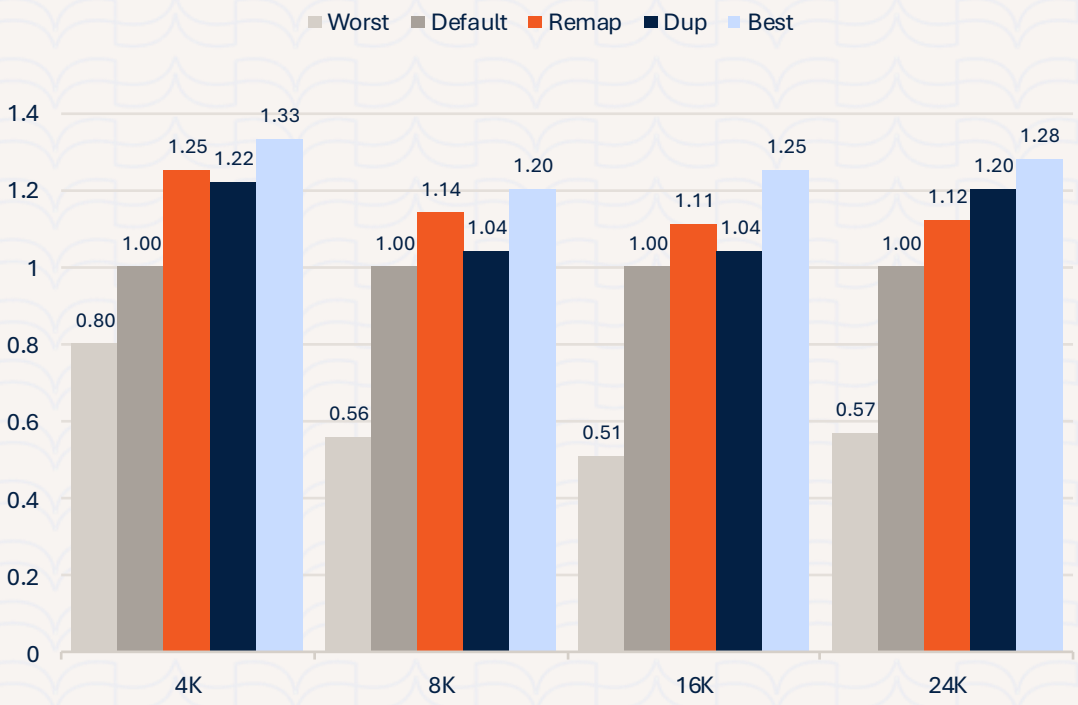


- **SGLang + DeepEP, EP8:** 16 experts per GPU per layer (17 with Dup); placement set through init_expert_location
- **Metric:** time of the three expert linear layers only; attention and all-to-all excluded
- **Best / Worst:** computed with oracle decode routing; bounds, **not practical options**

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Result: +12–15% on average, up to 1.25×

MOE COMPUTE SPEEDUP OVER DEFAULT



+15.5%
Remap avg

+12.5%
Dup avg

Remap averages within 9% of the oracle Best.

Placement matters even at EP8: Best is 1.7–2.5× Worst.

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Appx

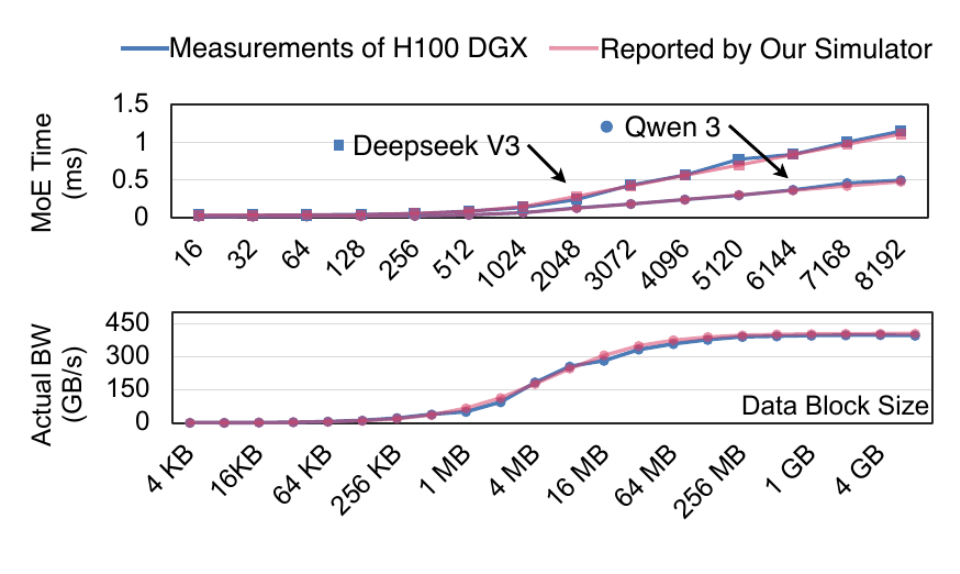
B1 · Full Fig. 12 throughput table

NORMALISED TO BASE; 5×5 DOJO | 8×3 TSMC-SOW

Model / variant	4096	8192	16384	4096	8192	16384
DeepSeek EP	7.9	5.6	3.1	10.2	7.2	3.9
DeepSeek Allo+Pred	8.1	6.6	3.9	10.3	8.4	5.0
Kimi EP	8.2	6.9	4.3	10.4	8.7	5.4
Kimi Allo+Pred	8.3	7.6	5.8	10.5	9.6	7.2
Llama EP	6.2	5.4	3.3	7.8	7.0	4.2
Llama Allo+Pred	6.7	6.8	5.8	8.5	8.5	7.2
Qwen EP	3.7	2.4	1.5	5.0	3.2	1.8
Qwen Allo+Pred	5.4	3.5	2.1	6.9	4.5	2.6

B2 · Is the simulator trustworthy?

PAPER FIG. 13: VALIDATION AGAINST A REAL DGX H100



- **Flat to ≈512 tokens:** weight read, 26 µs for an 88 MB expert
- **Then linear:** 1.15 ms at 8192 tokens ≈ 63% of peak
- **P2P:** saturates ≈400 GB/s above 64 MB
- **Not validated:** multi-hop mesh contention

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B3 · How can a wafer-sized chip yield?

CEREBRAS'S ARGUMENT, $D_0 \approx 0.001$ PER MM^2

≈360 mm^2

lost per GPU wafer

≈59 defects × one ≈6.2 mm^2 SM each

≈2.3 mm^2

lost per WSE-3 wafer

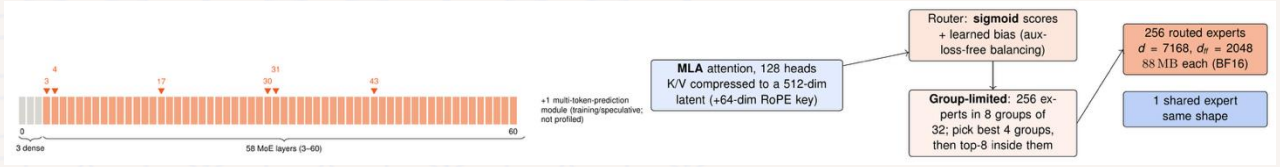
≈46 defects × one ≈0.05 mm^2 core each

≈100× less area lost per defect. Tiny redundant cores plus a fabric that routes around them; spares replace dead cores.

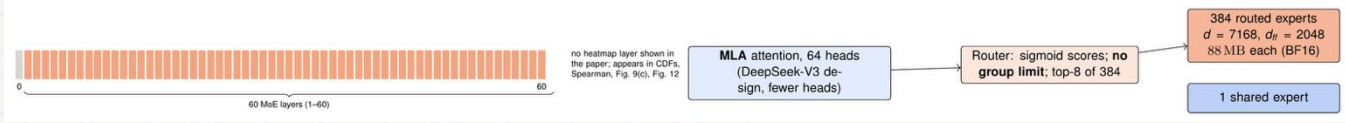
B4 · The four architectures

LAYER STACKS; ORANGE = MOE, GREY = DENSE; MARKERS = LAYERS IN PAPER FIGURES

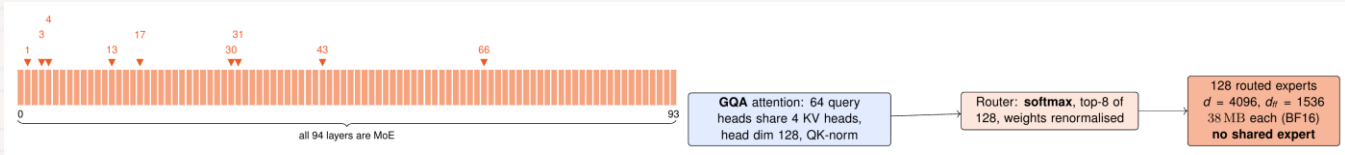
DeepSeek-V3



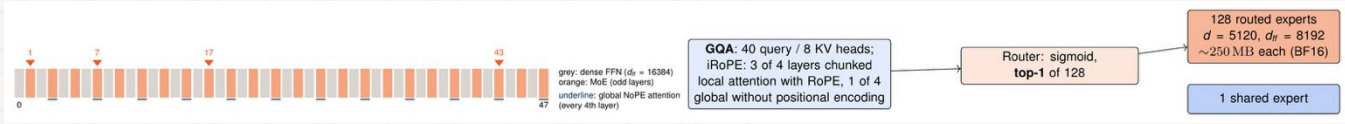
Kimi K2



Qwen3-235B



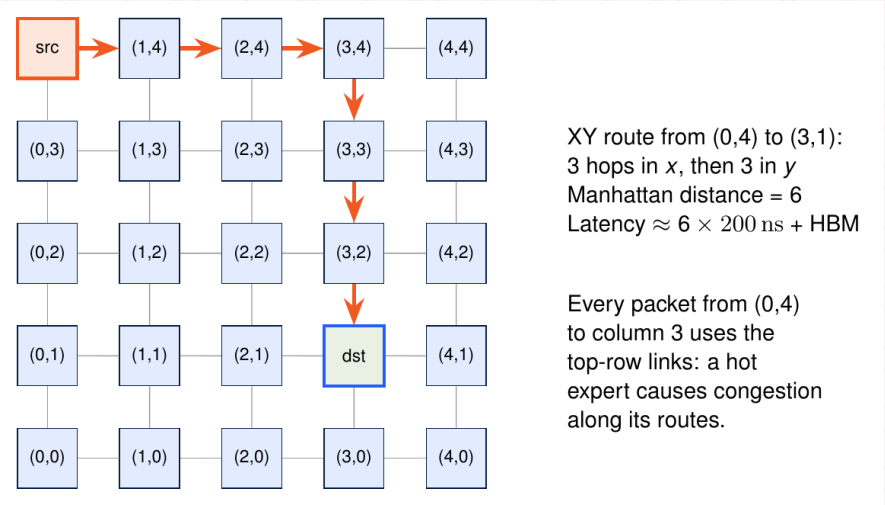
Llama 4 Maverick



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B8 · Why the 3×8 rectangle gains more

MESH GEOMETRY



XY route from (0,4) to (3,1):
3 hops in *x*, then 3 in *y*
Manhattan distance = 6
Latency $\approx 6 \times 200\,\text{ns}$ + HBM

Every packet from (0,4) to column 3 uses the top-row links: a hot expert causes congestion along its routes.

	5×5	3×8
Dies	25	24
Mean hops (random pair)	3.20	3.51
Max hops	8	9
Bisection links	5	3

Mean 1-D distance in a k-wide mesh = $(k^2 - 1) / 3k$.